CHEMICAL FORMULAS & NOMENCLATURE

General Chemistry I, Lecture Series 2

Qixi Mi

Reading:
OGB8 §1.2-3, §1.6, §2.1-2, §3.12-13

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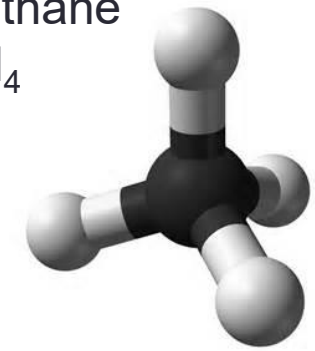
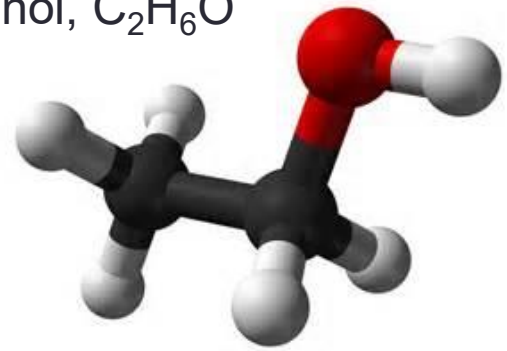
Outline

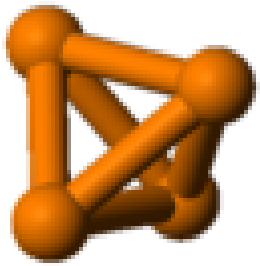
- Chemical formulas
  - Element, compound
  - Atom, molecule
  - The mole
  - Chemical equations
- Chemical nomenclature
  - Conventional
  - Systematic

Methane
CH_4

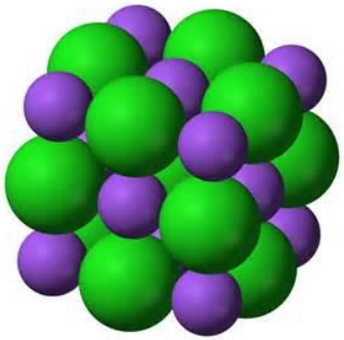


Ethanol, C_2H_6O

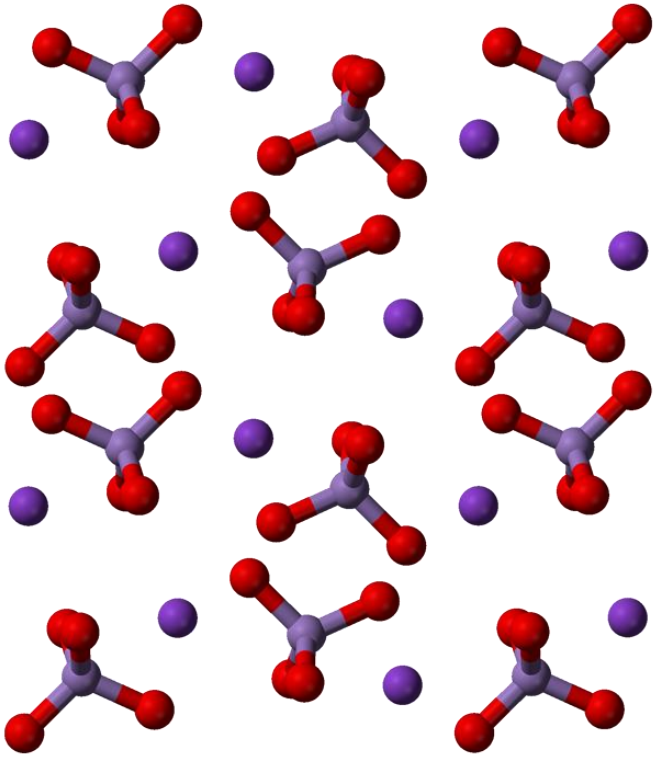




White phosphorus, P_4



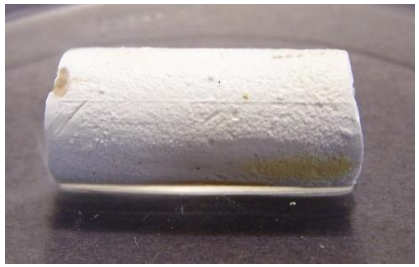
Sodium chloride, NaCl



Potassium permanganate, $KMnO_4$

Methane
CH_4

Ethanol, C_2H_6O



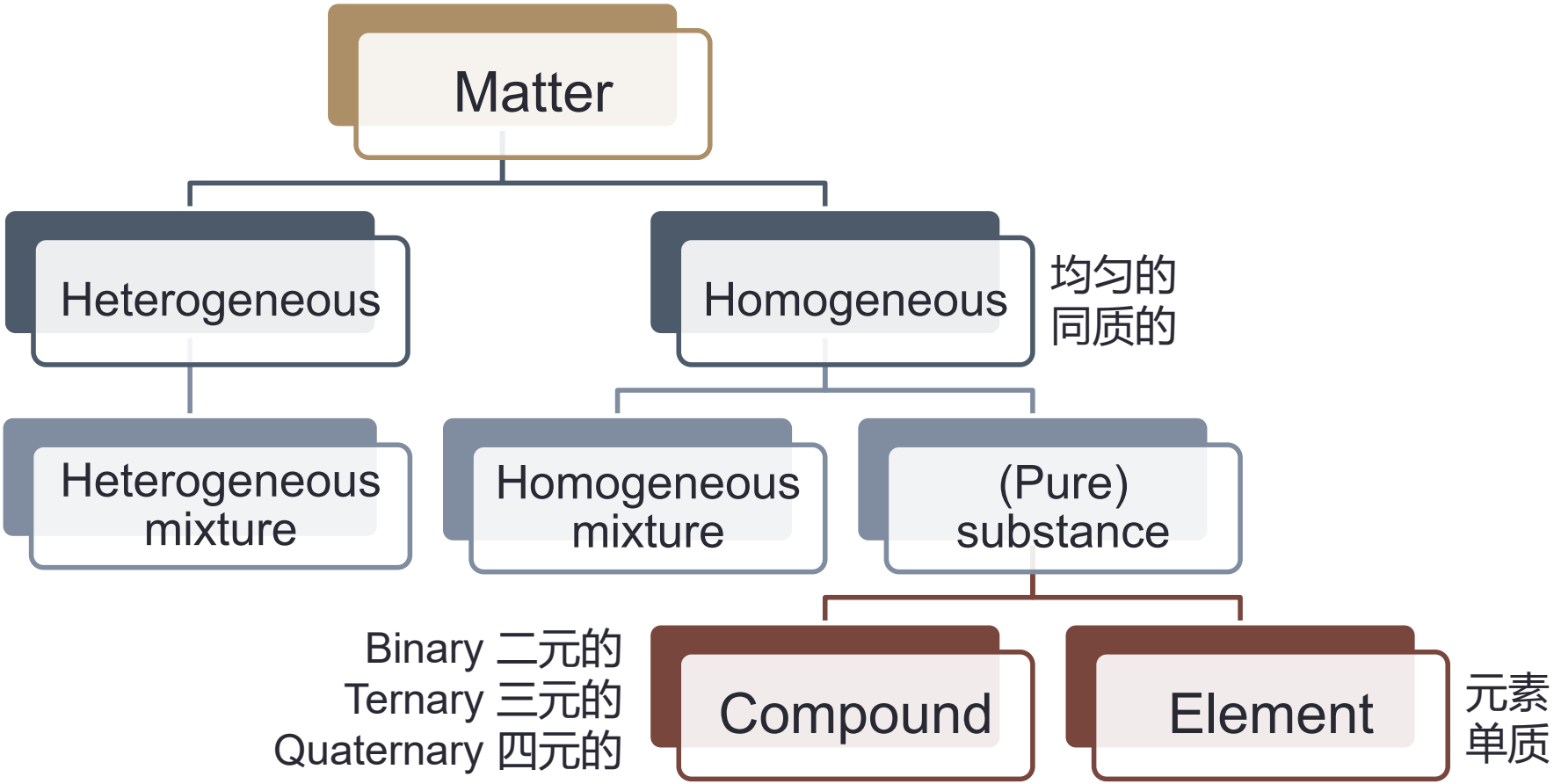
White phosphorus, P_4

Sodium chloride, NaCl

Potassium permanganate, $KMnO_4$

How were chemical formulas determined 100+ years before atoms could be seen?

Classification of Matter



Reductionism 还原主义 *vs.* Holism 整体主义

Refining Platinum

	1	2
5	4	3

Pt

Pt

$(NH_4)_2PtCl_6$

Chemical Elements

Atomic number

Symbol

Relative atomic mass

Platine
Platin
铂
白金

Name(s)

More Elements

Symbol	Names (EN, ZH, JA)	Meaning
H	Hydrogen, 氢, 水素	Water former
O	Oxygen, 氧, 酸素	Acid former
Cl	Chlorine, 氯, 塩素	Pale green
Hg	Mercury, Hydrargyrum, 汞, 水銀	Aqueous silver
Al	Aluminum, Aluminium, 铝, アルミニウム	Alum
W	Tungsten, Wolfram, 钨, タングステン	Heavy stone
Zn	Zinc, 锌, 亜鉛	–
Sn	Tin, 锡, スズ	–

Discovery of Oxygen

XXXVIII. *An Account of further Discoveries in Air. By the Rev.* Joseph Priestley, *LL.D. F. R. S. in Letters to Sir* John Pringle, *Bart. P. R. S. and the Rev. Dr.* Price, *F. R. S.*

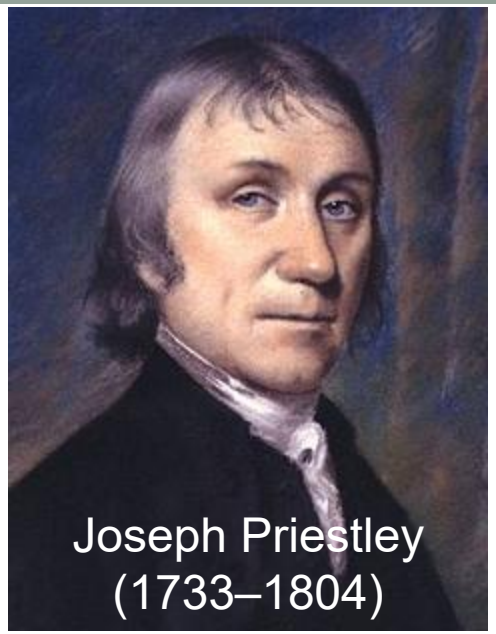
L E T T E R I.

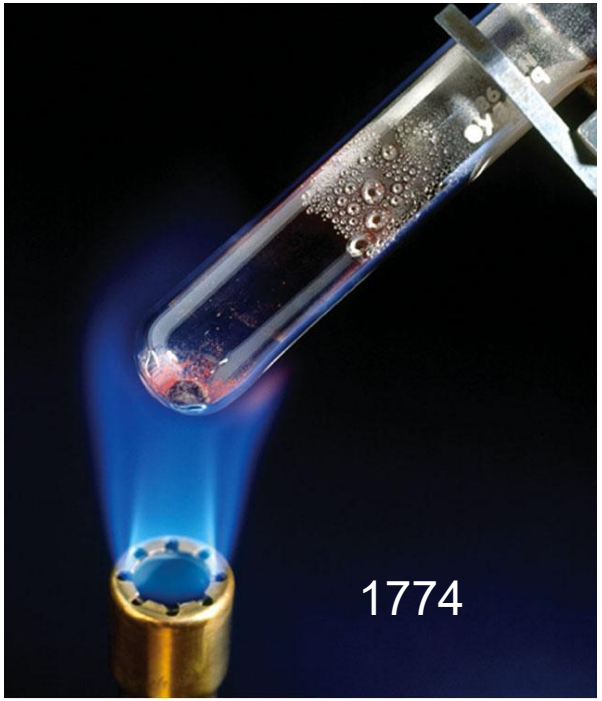
TO SIR JOHN PRINGLE, BART. P. R. S.

DEAR SIR, March 15, 1775.

Redde, May 25, 1775.

HAVING been pretty fortunate in the profecution of my experiments on different kinds of air, fince the publication of my treatife





other of its forms. But the moſt remarkable of all the kinds of air that I have produced by this proceſs is, one that is five or ſix times better than common air, for the purpoſe of reſpiration, inflammation, and, I believe, every other uſe of common atmoſpherical air. As I think I have ſufficiently proved, that the fitneſs of air for reſpiration depends upon its capacity to receive the *phlogiſton* exhaled from the lungs, this ſpecies may not improperly be called, *dephlogiſticated air*. This ſpecies of air I firſt produced from *mercurius calcinatus per ſe*, then from the red precipitate of mercury, and now from red lead. The two former of the ſubſtances yield it pure;

Lavoisier's Experiment

$$\text{mercury+air} \xrightarrow{350\ ^\circ\text{C}} \text{mercury oxide + remaining gas}$$

$$\text{mercury oxide} \xrightarrow{>500\ ^\circ\text{C}} \text{mercury + gas}$$

Conclusions from observations and quantitative measurements

- Air is a homogeneous mixture
- Chemical reactions occurred
- Mass was conserved during these chemical reactions

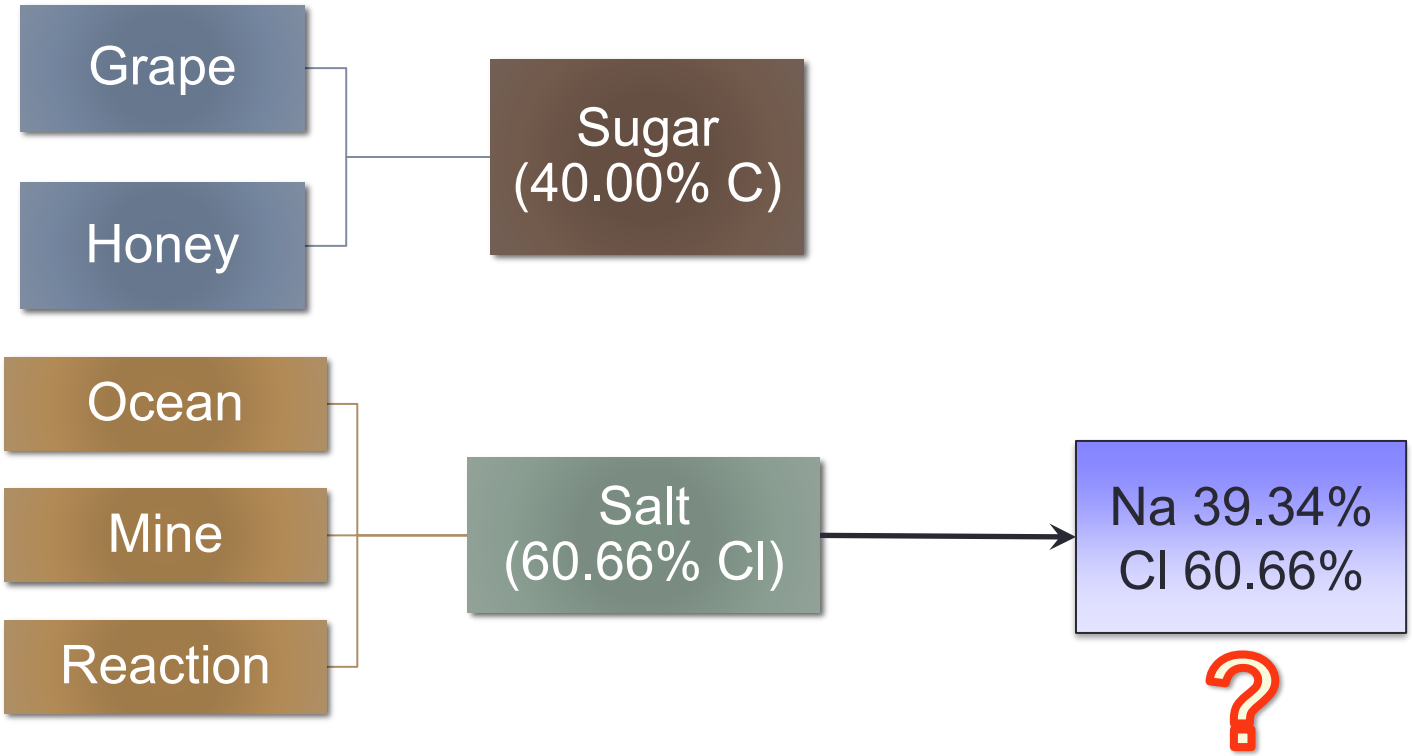
$$m(\text{mercury oxide}) = m(\text{mercury}) + m(\text{gas})$$

10.00 g 9.26 g 0.74 g

Antoine Lavoisier (1743–1794)

Stoichiometry
化学计量学

Compounds and Formulas

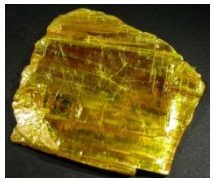


Law of Definite Proportions (1799) infers that (most) compounds

- Can and should be purified;
- Can and should be denoted by formulas.

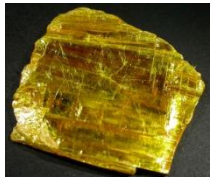
Law of Multiple Proportions

As: 70.03%
S: 29.97%



As: 60.91%
S: 39.09%

As / S
= 2.337



As / S
= 1.558

(As / S = 2.337) ÷ (As / S = 1.558) = 1.500 ≈ $\frac{3}{2}$

Formulas can and should be denoted by portions instead of compositions.

Dalton's Atomic Theory

Postulates atoms as portions of mass (1808).

- Atom
  - Identity — Element symbol
  - Quantity — Relative atomic mass (A_r)

John Dalton (1766–1844)

Compound	Experiment %
Hydrogen sulfide	H 5.92, S 94.08
Realgar	As 70.03, S 29.97
Orpiment	As 60.91, S 39.09

Dalton's Atomic Theory (2)

Assume: A_r(H) = 1, A_r(S) = 32, A_r(As) = 75.

Compound	Formula	Calculated %	Experiment %
Hydrogen sulfide	H_2S	H 5.88, S 94.12	H 5.92, S 94.08
Realgar	AsS	As 70.09, S 29.91	As 70.03, S 29.97
Orpiment	As_2S_3	As 60.98, S 39.02	As 60.91, S 39.09

Postulate 假定 → Hypothesis 假说 → Theory 理论

Empirical Formulas

For realgar "AsS", why not As_2S_2, As_3S_3, or As_4S_4?

For orpiment "As_2S_3", why not As_4S_6?

What is the formula of an element?

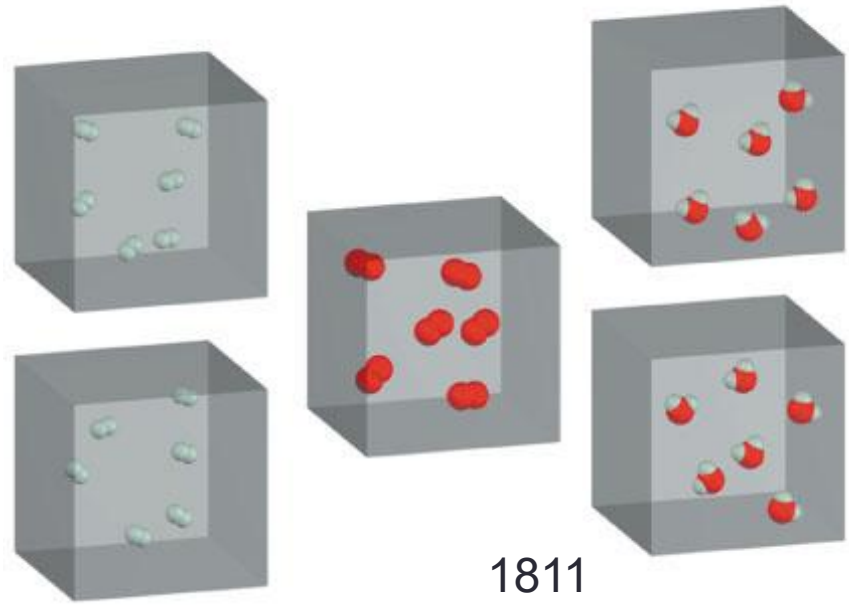
Hydrogen → H or H_2?

Oxygen → O, O_2, or O_3?

I prefer simplicity!

AsS and As_2S_3 are empirical formulas

Gaseous Reactions



1811

Joseph Louis Gay-Lussac
(1778–1850)

Hydrogen + Oxygen $\longrightarrow$ Water vapor

A_r	1	16	
Mass ratio	1	8	9
Volume ratio	2	1	2

Two independent sets of ratios

Avogadro's Hypothesis (1)

Dalton 1808: For all matter,

Number of atoms → Mass

Avogadro 1811: For all gases,

Number of molecules → Volume

Relative molecular mass M_r

Amedeo Avogadro
(1776–1856)

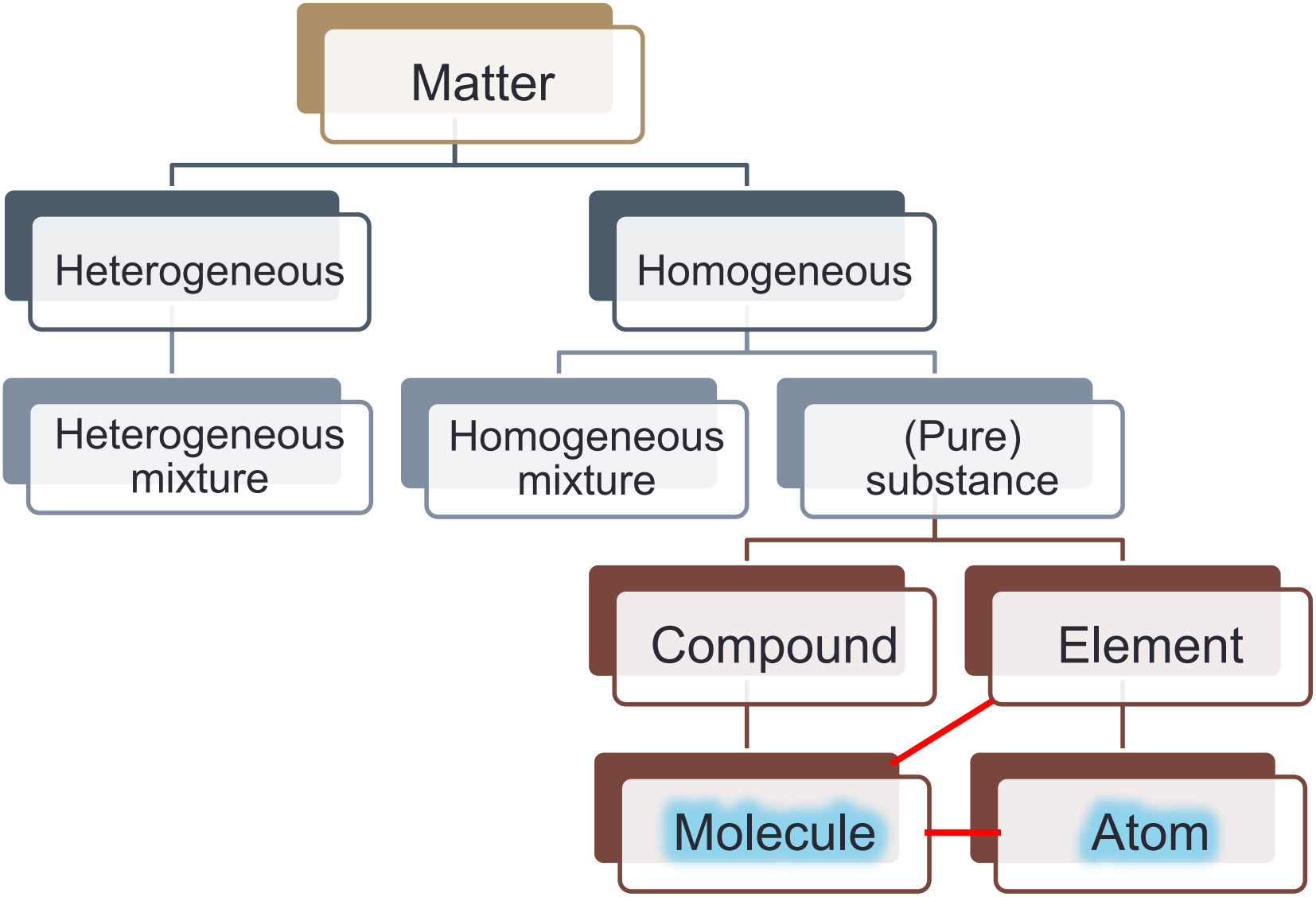
Avogadro's Hypothesis (2)

	Hydrogen +	Oxygen ⟶	Water vapor
A_r	1	16	
Mass ratio	1	8	9
Volume ratio	2	1	2
M_r ratio	0.5	8	4.5
M_r	1	16	9 (?)
M_r	2	32	18

$$2\,H + O \longrightarrow H_2O$$

OR

$$2\,H_2 + O_2 \longrightarrow 2\,H_2O$$
$$2\,H_4 + O_4 \longrightarrow 2\,H_4O_2$$


Matter
Heterogeneous
Homogeneous
Heterogeneous mixture
Homogeneous mixture
(Pure) substance
Compound
Element
Molecule
Atom

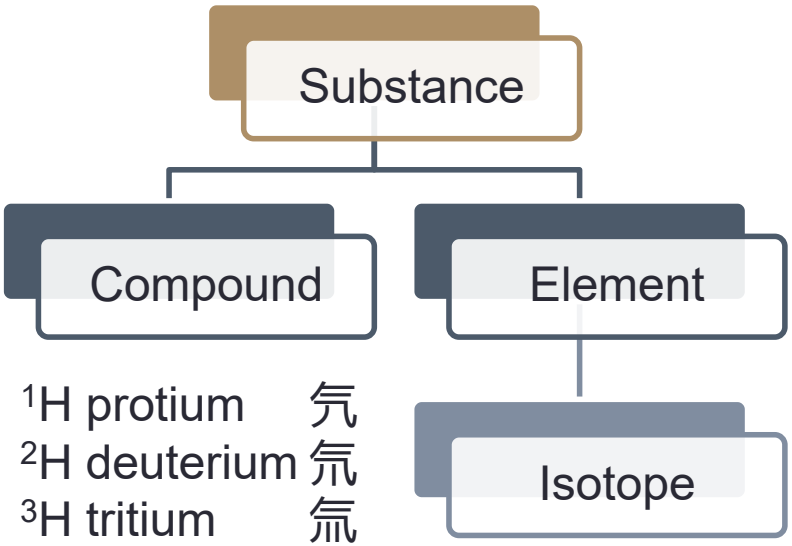
Outline

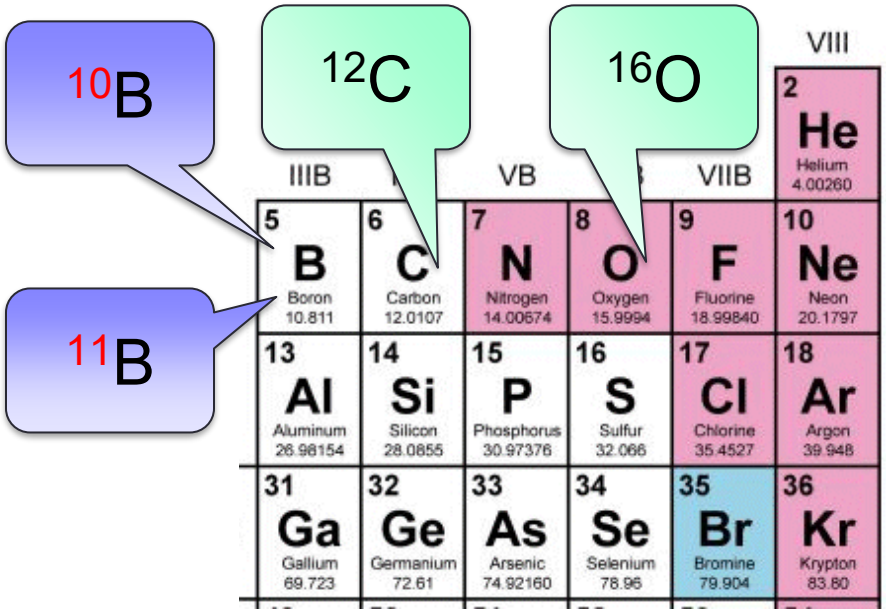
- Chemical formulas
  - Element, compound
  - Atom, molecule
  - The mole
  - Chemical equations

- Chemical nomenclature
  - Conventional
  - Systematic

Relative Atomic Mass and Isotope (1)

	H	B	C	O	Mg
A_r	1.0	10.8	12.0	16.0	24.3





Isotope 1	Abundance	Isotope 2	Abundance	A_r
^{10}B or B-10	~20%	^{11}B or B-11	~80%	B = 10.8
^{35}Cl or Cl-35	~75%	^{37}Cl or Cl-37	~25%	Cl = 35.5

Relative Atomic Mass and Isotope (2)

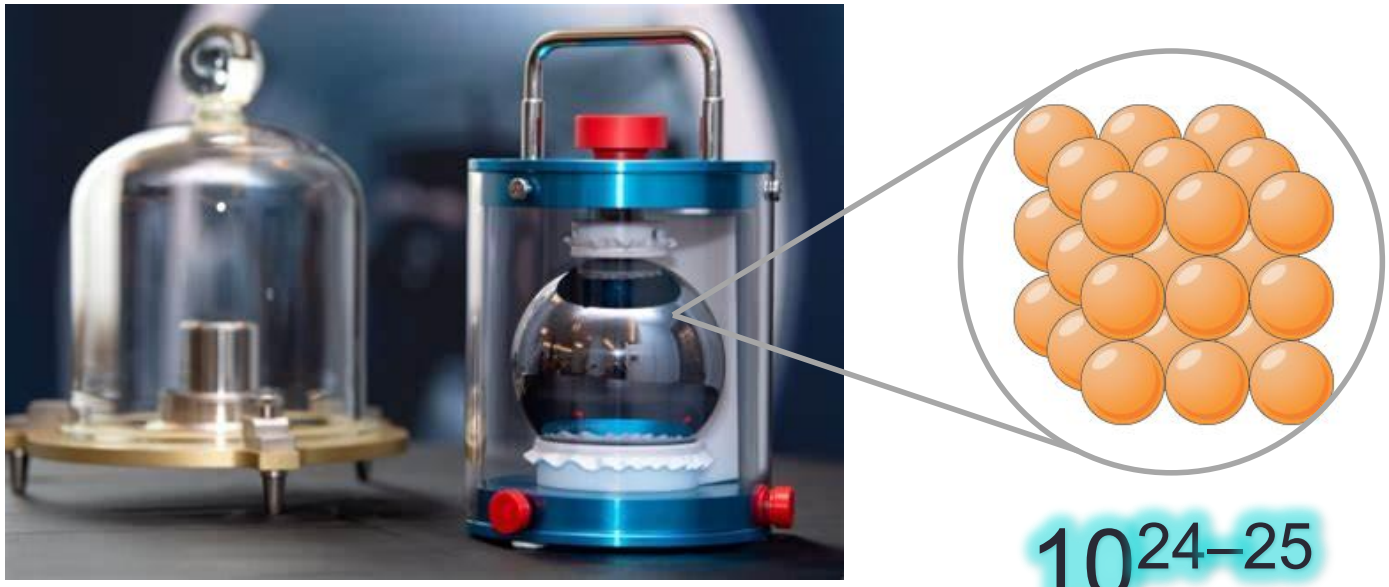
H	B	C	O	F
1.008 99.98%	10.013 80%	12 98.9%	15.995 99.76%	18.998 100%
2.014 0.02%	11.009 20%	13.003 1.1%	16.999 0.04%	
			17.999 0.2%	

1808 Dalton proposed $A_r(H) = 1$

~1900 Physicists took $A_r(O) = 16$

1961 Physicists & chemists agreed $A_r(^{12}C) = 12$

Absolute Atomic Mass (1)



1 kg 1 kg

$10^{24–25}$ atoms

Michael Faraday
(1791–1867)

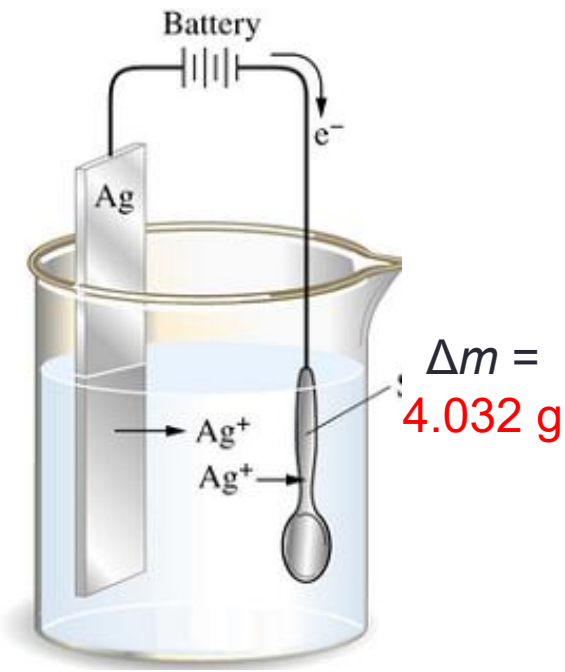
Robert A. Millikan
U. of Chicago / Caltech
(1868–1953)

1834 Faraday constant F = 96485 C/mol

1910 Millikan found $e = 1.60\times10^{-19}$ C

Absolute Atomic Mass (2)

1000 mA·h
(3600 C)



In 1834, Faraday measured 1.12 mg Ag per C

(Absolute) atomic mass m_a

$$m_a(\mathrm{Ag}) = 1.12\times10^{-3}\ \mathrm{g/C} \cdot 1.60\times10^{-19}\ \mathrm{C}$$
$$= 1.79\times10^{-22}\ \mathrm{g}$$

$$m_a(^{12}\mathrm{C}) = m_a(\mathrm{Ag}) \cdot \frac{A_r(^{12}\mathrm{C})}{A_r(\mathrm{Ag})}$$
$$= 1.99\times10^{-23}\ \mathrm{g}$$

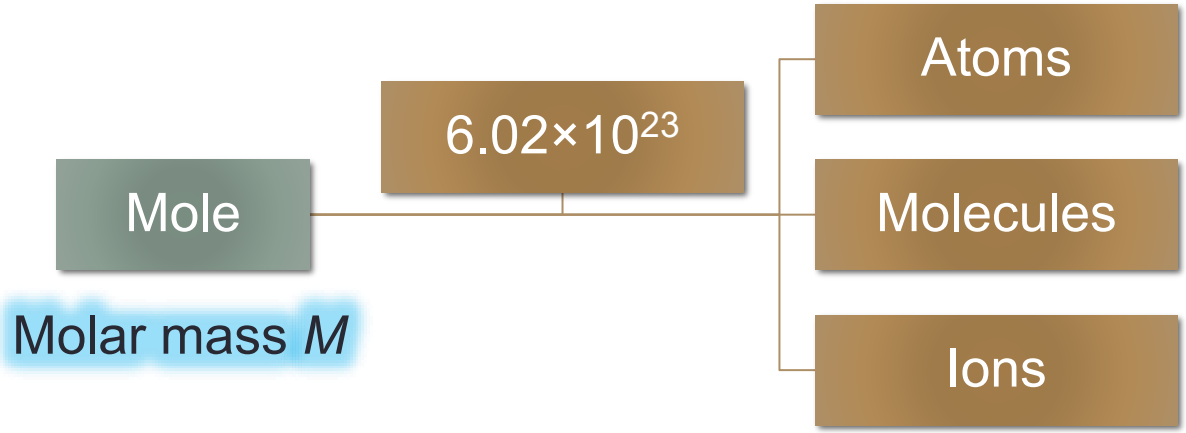
Electrochemistry

电化学

Avogadro's Number & the Mole

$$N_A \equiv \frac{12\ \text{g mol}^{-1}}{m_{a(^{12}C)}} = \frac{12\ \text{g mol}^{-1}}{1.99\times10^{-23}\ \text{g}} = 6.02\times10^{23}\ \text{mol}^{-1}.$$

$$N_A \equiv \frac{F}{e} = \frac{96485\ \text{C mol}^{-1}}{1.60\times10^{-19}\ \text{C}} = 6.02\times10^{23}\ \text{mol}^{-1}.$$



$A_r(\text{Ag}) = 107.9 \rightarrow M(\text{Ag}) = 107.9\ \text{g mol}^{-1}$.

$M_r(\text{H}_2\text{O}) = 18.0 \rightarrow M(\text{H}_2\text{O}) = 18.0\ \text{g mol}^{-1}$.

$M(\text{NaCl}) = 58.4\ \text{g mol}^{-1}$.

Chemical Stoichiometry

Review:

OGB8

§2.3 Writing Balanced Chemical Equations

§2.4 Mass Relationships in Chemical Reactions

§2.5 Limiting Reactant and Percentage Yield

Outline

- Chemical formulas
  - Element, compound
  - Atom, molecule
  - The mole
  - Chemical equations

- Chemical nomenclature
  - Conventional
  - Systematic

Simple Ions and Ionic Compounds

Cl^-	Chloride	氯离子
Na^+	Sodium ion	钠离子
NaCl	Sodium Chloride	氯化钠

DE: Natriumchlorid
FR: Le chlorure de sodium
JA: 塩化ナトリウム

NH_3	Ammonia	氨
NH_4^+	Ammonium	铵根[离子]

O_2^{2-}	Peroxide	过氧根[离子]	H_2O_2	Hydrogen peroxide
ClO^-	Hypochlorite	次氯酸根	HClO	Hypochlorous acid
ClO_2^-	chlorite	亚氯酸根	$HClO_2$	Chlorous acid
ClO_3^-	chlorate	氯酸根	$HClO_3$	Chloric acid
ClO_4^-	Perchlorate	高氯酸根	$HClO_4$	Perchloric acid

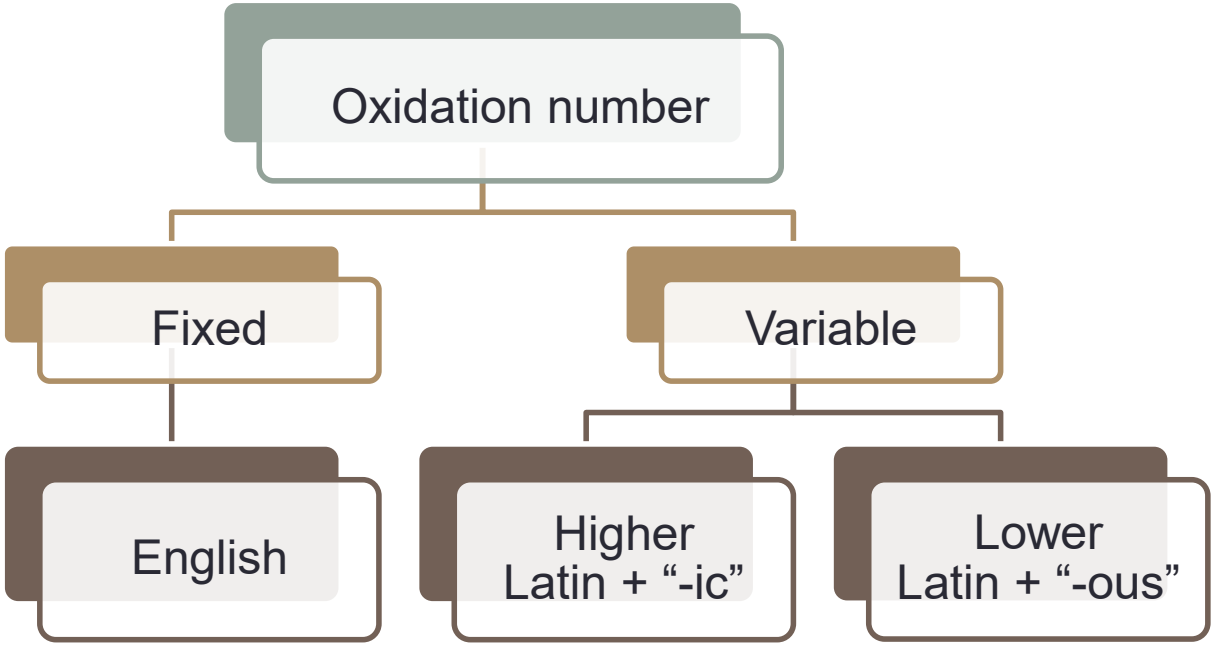
Greek Numerals in Names

HCO_3^- Hydrogen carbonate 碳酸氢根 OR bicarbonate

mono- 单	di- 二	tri- 三	tetra- 四	penta- 五
hexa- 六	hepta- 七	octa- 八	nona- 九	deca- 十

CO	Carbon monoxide	一氧化碳	mono- + -oxide = monoxide
CO_2	Carbon dioxide	二氧化碳	
$H_2PO_4^-$	Dihydrogen phosphate	磷酸二氢根	
$Cr_2O_7^{2-}$	Dichromate	重铬酸根	
As_2S_3	(Di)arsenic trisulfide	三硫化二砷	
As_4S_4	Tetraarsenic tetrasulfide	四硫化四砷	
N_2O_4	(Di)nitrogen tetroxide	四氧化二氮	tetra- + -oxide = tetroxide
P_2O_5	(Di)phosphorus pentoxide	五氧化二磷	penta- + -oxide = pentoxide
SF_6	Sulfur hexafluoride	六氟化硫	

Names of Metals



$Ca(OH)_2$	Calcium hydroxide	氢氧化钙
Al_2O_3	Aluminum oxide	氧化铝
AgBr	Silver bromide	溴化银

$FeCl_3$	Ferric chloride	(三)氯化铁
$FeCl_2$	Ferrous chloride	氯化亚铁
CuO	Cupric oxide	氧化铜
Cu_2O	Cuprous oxide	氧化亚铜
$SnCl_4$	Stannic chloride	(四)氯化锡
$SnCl_2$	Stannous chloride	氯化亚锡

Oxidation Numbers: Rules

Rule 1 Oxidation numbers sum up to the total charge.

Rule 2 H, Li, Na, K … = +1; F, Cl, Br … = −1; O = −2.

Rule 3 Elements on the lower left corner prefer a (+) oxidation number;
Elements on the upper right corner prefer a (−) oxidation number.

+1	+2											+3	−4 +4	−3 +5	−2 +6	−1 +7	
H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn

Only one common oxidation number

Oxidation Numbers: Examples

O_3	O(0)	LiH	Li(+1), H(−1)
N_2O	O(−2), N(+1)	ClO_4^-	O(−2), Cl(+7)
HSO_4^-	O(−2), H(+1), S(+6)	OF_2	F(−1), O(+2)
CuI	I(−1), Cu(+1)		

+1	+2											+3	−4 +4	−3 +5	−2 +6	−1 +7	
H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn

Only one common oxidation number

Systematic Nomenclature

O_3	Trioxygen
N_2O	Dinitrogen oxide OR Nitrogen(I) oxide
HSO_4^-	Hydrogen sulfate
CuI	Copper(I) iodide

Roman numerals for the oxidation number when it is ambiguous.

$CuSO_4 \cdot 5H_2O$	Copper(II) sulfate pentahydrate	五水合硫酸铜(II)
$LiFePO_4$	Lithium iron(II) phosphate	磷酸铁(II)锂
Hg_2Cl_2	Mercury(I) chloride OR Dimercury dichloride	

HBr	Hydrogen bromide	溴化氢	Hydrobromic acid 氢溴酸
HIO_4	Iodic(VII) acid	碘(VII)酸	[Periodic acid 高碘酸]
P_4	Tetraphosphorus	四磷	
D_2O	Deuterium oxide	氧化氘	

Periodic or Periodic?

Bottom Line

- These rules are made by chemists and for chemists.
- These rules are subject to revisions.
- The purpose of learning these rules is not only to abide by them, but, more importantly, to create new ones.